COMSATS University Islamabad

Attock Campus

Department Of Computer Science

Course	**Information Security**
Instructor	**Mam Ambreen Gull**
Assignment No.	**Lab 10**

Student Details

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Task 01:

```
def gcd(a, b):
    while b != 0:
        a, b = b, a % b    # swap and take remainder
    return a

# Example
print(gcd(48, 18))    # Output: 6
```

Output:

```
6


Process finished with exit co
```

Task 02:

```
def gcd_steps(a, b):
    while b != 0:
        print("a =", a, ", b =", b)
        a, b = b, a % b
    print("Final GCD =", a)
    return a

# Example
gcd_steps(48, 18)
```

Output:

```
a = 48 , b = 18
a = 18 , b = 12
a = 12 , b = 6
Final GCD = 6


Process finished with exit code 0
```

Task 03:

```python
def gcd_recursive(a, b):
    if b == 0:
        return a
    return gcd_recursive(b, a % b)

# Example
print(gcd_recursive(48, 18))    # Output: 6
```

Output:

```
Remainder of file ignored
6

Process finished with exit code 0
```